

NeuroTrade AI

User Guide — Quick Start

The main autonomous engine, the NeuroAI Quantum FAB, the five AI Bots, manual trading and every menu

Written against the code and live defaults in this repository · 3 September 2026

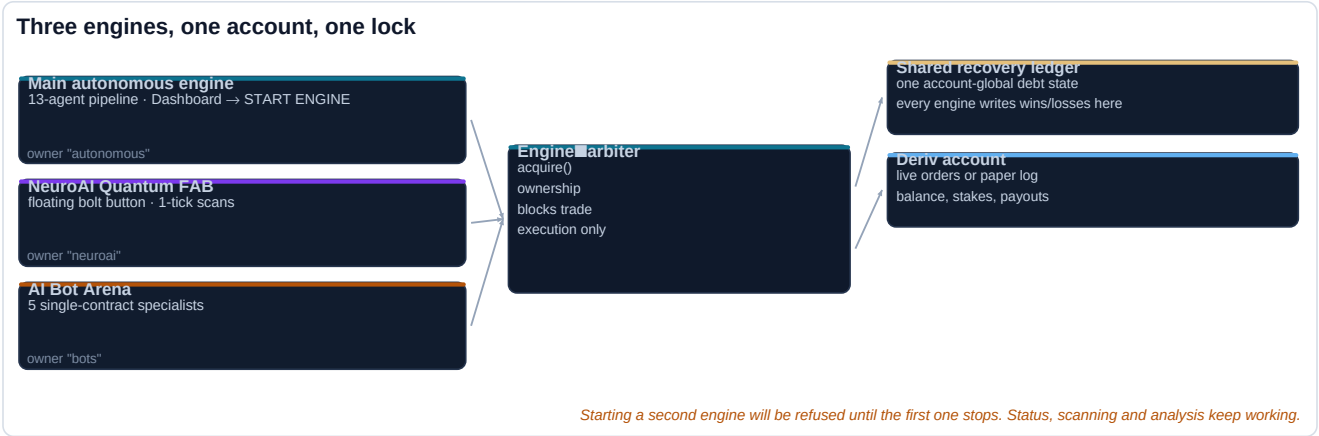
What this app is

NeuroTrade AI places and manages synthetic-index trades (Deriv volatility and jump indices) through three selectable engines and one journal. It is not a charting platform: every screen exists to answer one question — **is there a statistical edge in this market right now, and how big a stake does that edge justify?**

You can let the app trade by itself, run one of the specialist bots, drive the Quantum FAB, or trade manually with the AI telling you which side it favours. All four paths write to the same journal and the same recovery ledger.

Figure 1 — how the three engines share one account

Schematic of the live UI — not a screenshot. Drop a PNG at docs/guide/shots/architecture.png and rebuild to replace it.



Start here — the first-run checklist

1. Open **Connect** and attach a Deriv account. OAuth2 (the primary button) or a pasted Personal Access Token both work.
2. Confirm you are on a **Demo** account, not **Real**. The badge next to your login ID says which.
3. Go to **Settings** → **Engine Configuration** and turn **Paper Trade Mode ON**. It ships switched off.
4. Set **Risk Amount**, **Daily Loss Limit** and **Max Stake Per Trade** to numbers that suit the balance you connected.
5. Run the **Dashboard** → **START ENGINE** once, watch a few scans, then stop it and try the bolt-button FAB and one bot.
6. Only after you trust the settings: switch to a real account, and change at most one of those five things at a time.

One engine at a time

Only one engine can place trades at a time. If the autonomous engine is running, the FAB and the bots will refuse to start until it stops — and vice versa. This is deliberate: recovery debt is account-level, and two engines trading one account against two private recovery states produced real accounting failures before the lock existed.

In this guide

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1 · First run, and what the menus are

Before an account exists the app shows a landing page. That is a gate, not a bug: it keeps a new visitor out of a half-connected dashboard. Press the entry button and you land on **/connect**. Once an account is attached the normal interface opens on every route.

Connecting an account (Connect)

- **OAuth2 login (primary).** The main button starts an OAuth2 + PKCE flow against Deriv and returns to `/connect?code=...`. It asks for no secrets from you directly.
- **Bearer / Personal Access Token (fallback).** Paste an OAuth bearer token or a Deriv PAT to connect without the redirect. Useful when the OAuth app is not configured (the app needs `DERIV_APP_ID` for that path).
- **Demo vs Real.** Every account row carries an **Demo** (virtual) or **Real** badge, and the account switcher in the header lets you move between them — switching is a live action, so it is the one click to be deliberate about.

Do this before you start any engine

Paper Trade Mode defaults to OFF (Settings → Engine Configuration). With it off, an engine that is running sends live orders the moment it finds an entry it likes. Turn it on first: trades are then journalled exactly like real ones, with no order sent to Deriv.

Where the numbers come from

Ticks arrive over Deriv's public WebSocket. If that socket cannot connect, the app does not go blank — it starts a simulated price feed and stamps the dashboard header with **SIM DATA**. Trades are still recorded, and in simulation their win/loss outcome is weighted by the AI's own confidence rather than a settled contract. So a session showing **SIM DATA** tells you how the engine behaves, not how the market treated you.

The nine menus

Menu	What it is	Reach for it when
Dashboard	The main autonomous engine: status, 13 agents, performance today, recovery state, daily target.	You want the app to run itself, or to check on it.
Markets	Ranked list of the synthetic markets with quality, confidence, risk and trend scores.	You want to pick or inspect an instrument.
AI Bots	The AI Bot Arena: five single-contract specialists with their own deploy consoles.	You want one trade type handled by an expert model.
Journal	Trade-by-trade record of what was placed, on which contract, at what stake, and the result.	You are reviewing what actually happened.
Analytics	All-time performance: win rate, payout mix, market-level breakdowns.	You want history rather than today.
Intelligence	Post-trade learning: missed opportunities, threshold tuning, generated reports.	You want to know what the app learned and what it skipped.
Risk Calc	Standalone sizing calculator — stake, recovery steps, exposure.	Before you commit to a multiplier or a stake.
Settings	Risk, daily limits, engine behaviour, recovery mode, barriers, contract mode.	Any time behaviour needs changing.
Connect	Account connection, tokens, Demo/Real switching.	Setup and account changes.

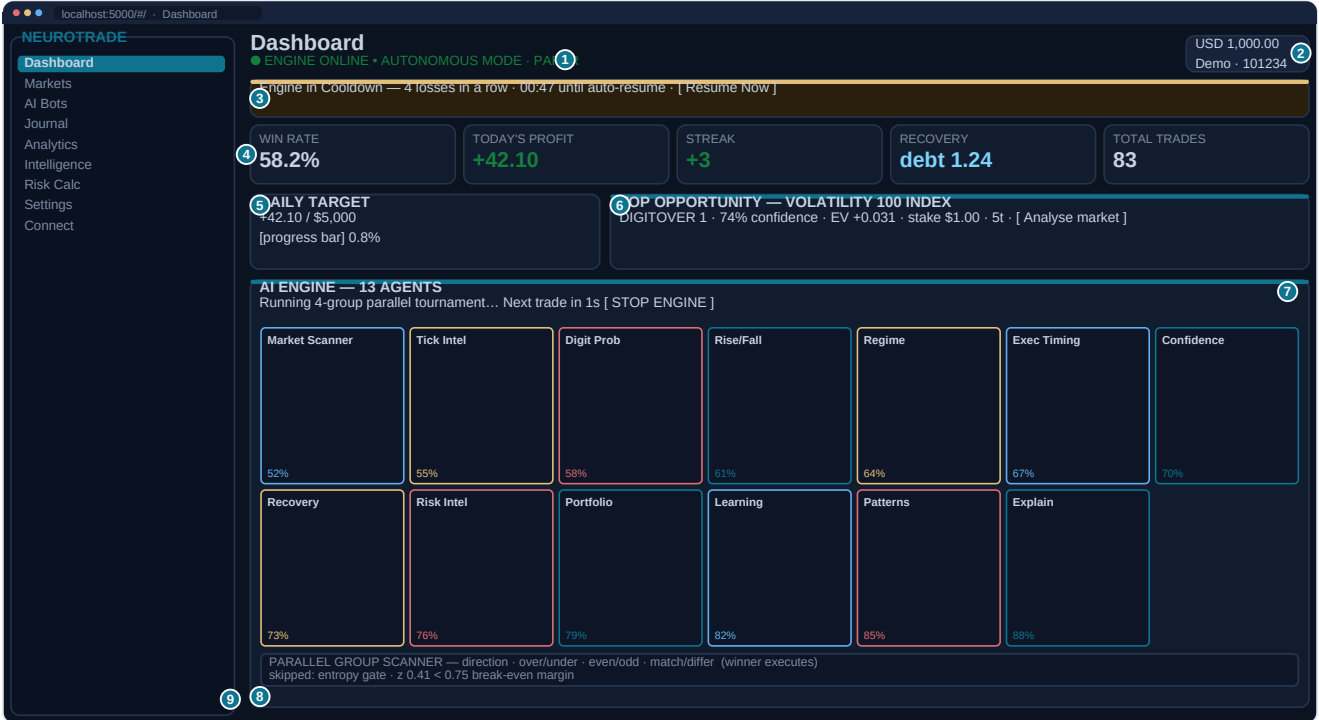
Everything below assumes you are on a Demo account with Paper Trade Mode on.

2 · The Dashboard, region by region

The dashboard is the main engine's cockpit. Nothing on it is decorative — every block is either an output of the 13-agent pipeline or a safety switch.

Figure 2 — Dashboard layout (the numbered markers key to the table below)

Schematic of the live UI — not a screenshot. Drop a PNG at docs/guide/shots/dashboard.png and rebuild to replace it.



#	Region	What it tells you
1	Status line	ENGINE ONLINE / ENGINE STANDBY / COOLDOWN, then AUTONOMOUS or MANUAL mode, plus PAPER and SIM DATA tags when those apply. Read this first, every time.
2	Account chip	Login shown as currency and balance. If it is absent the app has no account and the engine has nothing to trade.
3	Cooldown banner	Appears after the consecutive-loss limit, with the stop reason and a countdown to auto-resume. Resume Now restarts early.
4	Performance strip	Today's win rate, profit, streak, recovery state and trade count. Day-scoped on purpose — all-time numbers live in Analytics.
5	Daily target	Progress toward the profit target with the limit that halts trading. If the loss limit is hit you get Loss limit hit — trading paused.
6	Top opportunity	The best current setup as a 3D flash card: market, contract, confidence, EV, suggested stake and duration.
7	START / STOP ENGINE	The master switch for the autonomous engine, with today's executed-trade count next to it.
8	Scanner + agent grid	The four contract groups racing for the next trade, then each agent's live confidence.
9	Navigation rail	The nine menus. Kept highlighted for the section you are in.

Reading the agent grid

Thirteen tiles, coloured by their own confidence: **70 and above** is calm green, **50–70** is amber, **below 50** is red. A single red tile is normal — the pipeline is a consensus, not a veto — but a floor of reds means the engine is being held back by the tape, not by your settings.

3 · Running the main autonomous engine

1. **Press START ENGINE.** The status line flips to ENGINE ONLINE • AUTONOMOUS MODE and the scan bar appears.
2. **Watch one tournament.** The bar cycles *Scanning markets...* → *Running 4-group parallel tournament...* → *Executing: <group>*, with *Next trade in Ns* driven by your Scan Interval (default 1 s).
3. **Let it skip.** Most loops end in no trade. The scanner line and the red “skipped” reasons are the product working, not failing — with *Require Positive EV* on, no entry is good enough far more often than a live trader expects.
4. **Check the journal, not the hype.** Trades land in the Journal with contract, stake, payout and result. Today’s roll-up is in the performance strip.
5. **Stop it with STOP ENGINE** when you want the machine back. Ownership of the account is released, so the FAB or a bot can then start.

What the 13 agents actually do

Each scan runs a staged pipeline. Agents that can work in parallel do; the decision at the end consumes all of them.

Stage	Agents	Job, in plain words
1	Feature Engineering	Turns the raw tick stream into the windows, streaks and digit matrices everything else reads.
2	Market Scanner · Tick Intelligence · Market Regime	Is this market worth trading, what are the last ticks really doing, and is the tape trending, choppy or violent?
3	Digit Probability · Rise/Fall · Portfolio Manager · Recovery Intelligence	Per-contract maths: digit and direction odds, how a stake fits the account, and what the open recovery debt demands.
3.5	Duration Optimizer	Chooses the tick count for the contract rather than always using the default.
4	EV Calculator	Converts probability and payout into expected value, and that into a stake.
5	Risk Intelligence · Execution Timing · Learning Agent	Circuit-breakers, whether this instant is the right instant, and what past results say.
6	Pattern Discovery · Confidence Fusion	Recurring shapes in the tape, then one weighted confidence number from every opinion above.
7	Trade Explainability	The human-readable “why this trade”, shown on the card and in the journal.
8	Master Decision	Trade or pass — the only component that fires.

Limits that stop it for you

Setting (default)	Effect when reached
Daily Loss Limit (2,999)	Trading halts for the day; the dashboard says <i>Loss limit hit — trading paused</i> .
Daily Profit Target (5,000)	Stops on target. Note the default assumes a large account — lower it before a small one runs live.
Consecutive Loss Limit (4)	Cooldown starts, then auto-resume; the banner explains why.
Max Drawdown (10 %)	Stops if the portfolio drops by this percentage.
Min Confidence Threshold (50)	Any signal below this is discarded. The Intelligence page can raise it automatically from your realised win rate.
Max Stake Per Trade (500)	Hard cap per trade regardless of balance or recovery maths.

Recovery defaults

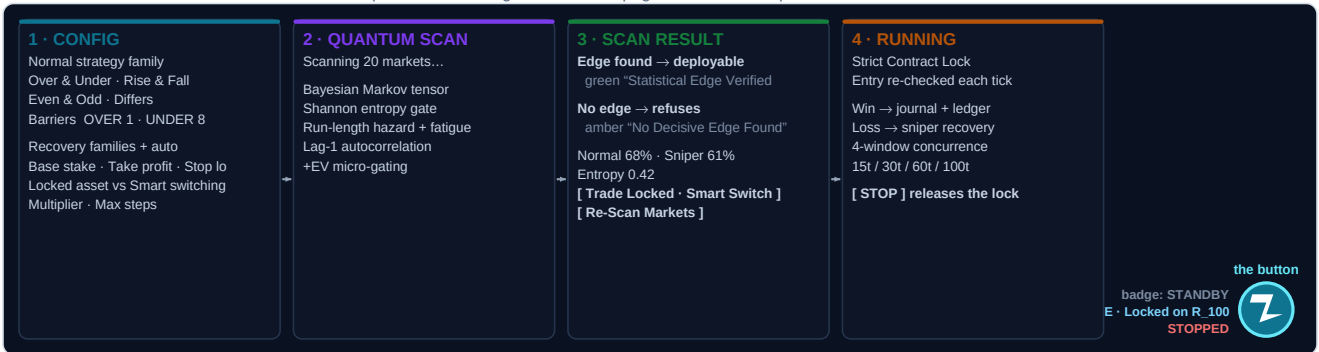
Recovery is **on by default** (Recovery Mode, multiplier 1.62, 3 steps). In **Auto** calculator mode the multiplier is not even read — the stake is recomputed from live debt and the live payout. Switch to **Manual** and that 1.62 becomes literal and compounding: ×1.62, then ×2.62, then ×4.23 of base stake. Do not flip that switch until you have watched a session in Auto.

4 · The NeuroAI Quantum FAB

The lightning-bolt button pinned to the bottom-right of every screen opens a self-contained fast engine — a separate, more aggressive strategy that scans, decides and executes on a one-tick rhythm. It keeps its own settings inside the panel (it does not read the Dashboard's engine settings for its own risk numbers) and it competes for the same single execution lock, so the main engine must be stopped first.

Figure 3 — the FAB panel through its four steps

Schematic of the live UI — not a screenshot. Drop a PNG at docs/guide/shots/fab.png and rebuild to replace it.



The controls, and what they change

Control	What it does
Normal Trade Strategy	Which contract family it hunts for ordinary trades: Over & Under (digit barriers), Rise & Fall (direction), Even & Odd (parity) or Differs (cold-digit avoidance, ~96 % win rate at a 1.09× payout).
Normal barriers	The OVER and UNDER digits when that family is armed. The default OVER 1 / UNDER 8 sits near an ~80 % hit rate for a small payout.
Sniper Recovery Strategy	Which families may be used while recovering, separately from the normal choice — including Matches at 8.93×, which is the fastest way out of a small debt and the fastest way to grow a big one.
Recovery auto mode	On = the stake is computed from live debt and live payout. Off = your multiplier, compounded per step.
Base stake / Take profit / Stop loss	Session risk for the FAB alone. Base stake has a 0.35 floor to stay above Deriv's minimum.
Market mode	Locked trades one chosen asset only; Smart switching lets the engine rotate to wherever the edge is.

How it decides — one paragraph, no marketing

The scan blends a second-order Bayesian Markov model over digit transitions (with a Dirichlet prior so a cold digit is not trusted on two appearances), Shannon entropy to reject noise-shaped streams, a geometric hazard model that asks whether a run of repeated digits is *due to break* rather than whether it is long, lag-1 autocorrelation and micro-tick acceleration for direction, and an expected-value gate that requires the blended probability to clear the contract's own break-even. Recovery adds a four-window concurrence check — the same edge has to appear over 15, 30, 60 and 100 ticks — before an escalated stake is allowed.

The scan result is allowed to say no

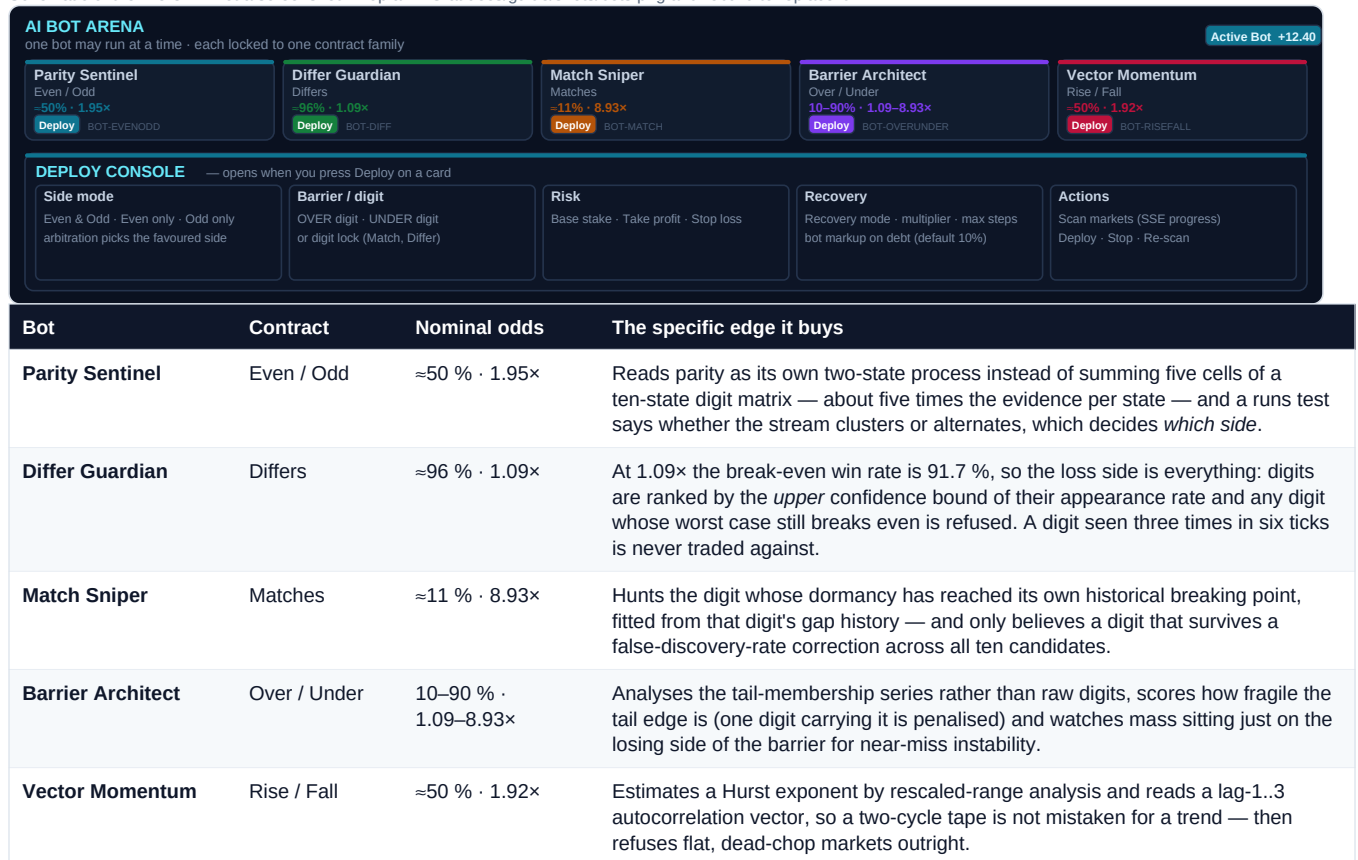
If the result screen says **No Decisive Edge Found**, the honest move is **Re-Scan** or walk away. Deploying anyway converts a statistical engine into a coin flip with a house margin.

5 · The AI Bots — five single-contract specialists

Each bot trades exactly one contract family and spends its whole analysis budget there. That is the entire premise: a generalist splits its statistics across six families, so a specialist gets sharper estimates for the one it owns — and the app copies the FAB's formulas rather than reusing its session, so a bot's numbers stay comparable to the FAB's without inheriting its execution.

Figure 4 — the arena and a deploy console

Schematic of the live UI — not a screenshot. Drop a PNG at docs/guide/shots/bots.png and rebuild to replace it.



Deploy console, in order

- 1. Side mode.** Bots with two sides offer *both* (analyse each, execute the favoured one) or lock to a single side. Digit bots offer a **digit lock**: auto-pick, or force one digit.
- 2. Barriers / digits.** Only shown where the family has them (OVER/UNDER digits for Barrier Architect).
- 3. Risk.** Base stake, take profit, stop loss — session-scoped, like the FAB.
- 4. Recovery.** Mode, multiplier, max steps, plus the bot's recovery markup on debt (default **10 %**): a recovery stake is sized so one win clears all debt *plus* that percentage as profit, instead of using the general recovery target that suits high-payout contracts badly.
- 5. Scan markets.** Scores the candidate markets live (progress streams while it works), then **Deploy** starts the session; **Stop** ends it and releases the lock.

Why a bot may do nothing for a long time

Every bot entry is measured against the contract's **break-even** win rate ($1 \div \text{payout}$), never against 50 %. A 60 % hit rate on a 1.09× Differs is a long-term loss. The gates are deliberately strict — parity and momentum need 0.75σ of margin over break-even, a 1-digit barrier tail needs 1.25σ , and Matches need 1.5σ because picking the best of ten digits flatters the estimate. Few, late, boring trades are the intended behaviour.

6 · Manual mode, Markets, and reading your results

Manual mode

The status line reads **MANUAL MODE** whenever no engine holds execution — that is the whole definition. Nothing is placed for you, but the analysis keeps running, which is the point: manual mode is for taking the machine's read and pulling the trigger yourself.

- 1. Open Markets** and pick a market (or press **Analyse market** on the dashboard's top-opportunity card).

2. On the market page, the contract buttons carry the pipeline's opinion directly: **AI FAVOURS** · **5t** on the favoured side, plus the recommended duration.
3. Clicking Rise/Fall or Even/Odd **auto-fills the stake and tick count** from the agent output, so you confirm a size rather than compute one.
4. The per-market panel shows signal strength and the AI signal sentence, and a manual trade is scored by the same assist path the engines use — *trades/assist* runs the Quantum timing on your chosen entry so the outcome lands in the same journal and the same recovery ledger.

Markets and the market page

The market list is sorted by the scanner's ranking, so the top row is where the engine is looking. Each row shows the quality, confidence and risk scores and the detected trend; opening a market gives the live tick stream, the digit distribution, and the contract-by-contract read. This is also where you sanity-check what an engine is doing before blaming the settings.

Journal · Analytics · Intelligence · Risk Calc

Screen	Use it for
Journal	The factual record: every trade with market, contract, stake, payout, result, which engine placed it, and whether it was part of a recovery sequence. Start any review here.
Analytics	All-time aggregates, so you can see whether today was noise or a trend — and separate performance by contract family and market.
Intelligence	What the app learned from your trades: missed opportunities it would have taken, the confidence threshold it recommends from realised win rate, and generated reports. This is the page that argues with your settings.
Risk Calc	Sizing arithmetic on its own: stake per trade and what a recovery ladder costs. Use it before you accept a multiplier, not after.

Choosing how to trade — the four paths side by side

	Autonomous engine	NeuroAI FAB	AI Bots	Manual
Who places the trade	The 13-agent pipeline, every scan interval	The bolt panel, tick by tick	One specialist, its own loop	You, from the market page
Contract choice	Any family you allow in Settings	One normal family you pick in the panel	Hard-locked to that bot's single family	Whatever you click
Stake source	Risk Amount / Percentage in Settings	Base stake in the panel	Base stake in the console	You set it, auto-filled on request
Recovery	Shared ledger, Auto calculator by default	Shared ledger, its own multiplier fields	Shared ledger + bot markup on debt	Recorded into the same ledger
Best used for	Hands-off sessions; the broadest market coverage	Fast, focused action on one family	Squeezing one contract type properly	Learning what the AI sees, at your own pace
Watch out for	It inherits the global Daily Limits	Its panel risk values are separate	May idle for a long time — by design	Nothing pauses for you; limits are yours

The only three settings that matter on day one

Whichever path you choose, the same three numbers decide whether a bad day stays survivable: **Daily Loss Limit**, **Max Stake Per Trade** and whether **Paper Trade Mode** is still on. Change one at a time, and give each change a full session before the next.

7 · Settings reference, then troubleshooting

The six cards, with the shipped defaults read from this install. Values are stored per browser session, which is why a fresh profile starts at these numbers.

Setting (card · field)	Default	What it does / when to touch it
Risk Profile · Profile preset	Moderate	Feeds the stake-sizing multiplier. A preset, not an override.
Risk Amount Type / Value	Fixed · 1.00	Per-trade risk as a fixed amount or a percentage of the connected balance.
Max Stake Per Trade	500	Hard cap per trade, regardless of balance or recovery maths.
Daily Limits · Profit Target / Loss Limit	5,000 / 2,999	Stop on target; halt for the day on limit. Both are scaled for a large account — reduce them first .
Max Drawdown · Consecutive Losses · Cooldown	10 % · 4 · 1 min	Portfolio stop, then a timed pause with a countdown and <i>Resume Now</i> .
Engine Config · Paper Trade Mode	OFF	Turn ON first. Off means an engine sends live orders the moment it finds an entry.
Require Positive EV · Min Confidence Threshold	ON · 50	The two quality dials. Raise the threshold for fewer, better trades; lower it for more noise.
Scan Interval · Trade Duration · Scan All Markets	1 s · 5 t · ON	How often it re-evaluates, default contract length (the Duration Optimizer may override), and market scope.
Recovery · Enable · Calculator mode	ON · Auto	One global, account-level debt state shared by all three engines. Auto sizes the stake from live debt and payout — the multiplier field is not read .
Recovery Method · Max Recovery Steps	Split · 3	Split never stakes more than one base stake per attempt and carries debt forward; Instant tries to clear it in one win.
Recovery Multiplier (Manual mode only)	1.62×	Used literally and compounded per step: $\times 1.62 \rightarrow \times 2.62 \rightarrow \times 4.23$ of base stake.
Barriers · Normal OVER / UNDER	1 / 8	$\approx 80\%$ hit rate for a small payout. Recovery uses OVER 3 / UNDER 6 with live proposals beating the fallback.
Contract Mode	All selected	All selected lets the AI pick the best contract per opportunity; Selective restricts it to your categories.

Troubleshooting

Symptom	What it means
Engine will not start; error mentions another engine	Another engine owns execution. Stop the FAB or the bot (or the main engine) and retry.
Header shows SIM DATA	No live tick socket. Prices are simulated; the UI works, but results are not market-settled.
“Connect Deriv Account” badge stays in the header	No account attached to this browser session, so nothing can trade.
Lots of “skipped”, no trades	Working as intended: EV and confidence gates. Lower <i>Min Confidence Threshold</i> only if you want more noise.
Cooldown banner right after starting	You are resuming into an open streak. Check the loss limit and the recovery settings before resuming again.
Trades look identical to a bad day last week	Recovery debt persists across engines. The recovery card on the dashboard can clear it deliberately.

Replacing these diagrams with real screenshots

The figures above are drawn schematics, labelled as such — they match the live layout but they are not captures of it. To swap in real screenshots, run the capture script from the repository root on a machine with a browser and the dev server up:

```
npm i -D playwright && npx playwright install chromium
node docs/guide/capture-screenshots.mjs
```

It writes docs/guide/shots/<name>.png for each screen. Re-run `python3 docs/guide/build_user_guide.py` and every diagram that has a matching PNG is replaced by that screenshot, with a green “● screenshot” tag so the guide always says which is which.

Figure slots in this edition: dashboard · fab · bots · architecture. The capture script also saves the other screens, ready for new figures.